Project by
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SOLAR MAP

Transforming Bahrain's Solar Energy Landscape with Cutting-edge Technology

Why Solar Map?

- Real-Time Solar Panel Visualization (Monitoring and Management)
- Precise Solar Panel Calculation (Cost & ROI Predictions)
- Intuitive and User-Friendly Comprehensive Platform

Introduction and Project Overview

Introduction

Today marks the launch of SolarMap, a collaboration between AWS, Tamkeen, BP, and MEWA in Bahrain. SolarMap estimates the cost and number of solar panels, manages and maps installations using AI for Rooftop Segementation, Solar Panel Detection and AWS Ground Station as a Service. This platform supports Bahrain's sustainable energy goals and streamlines solar management.

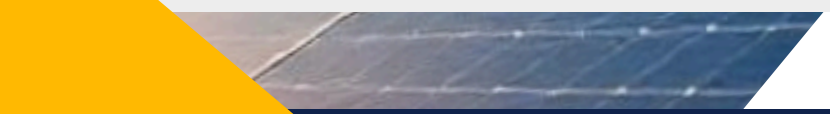
Current Challenges

The struggle to meet the growing demand for solar panels. Factors like residence size, location, and power usage complicate accurate estimations. Traditional methods, such as manual applications and reading electricity bills, are time-consuming and inefficient.



Innovative Solution

SolarMap's calculation capabilities allow for precise cost estimations and ROI calculations by extracting location data from electricity bills and determining the required number and cost of panels online. Developed with AWS, SolarMap uses the latest satellite technology and cloud computing for a comprehensive overview of solar installations.



Functionality and User Benefits



Why Solar Map?

User-Friendly Platform

Homeowners can use the web portal to access the Map/Calculation page, fill in their details (kilowatt usage, rooftop size, etc.), or streamline by uploading a electricity bills and instantly get cost and ROI estimations . They can then inquire further through form submissions or interactive chatbot and get real-time recommended contractors.

Administrative Control

The application administrators will have access to inquiries and can send emails to applicants. The admins can also visualize solar panel installations across Bahrain and aggregate future installations on a map and utilize the Business Intelligence Dashboards.

Real-Time Monitoring

SolarMap solves the time-consuming and costly process of manually mapping and managing solar installations. It offers real-time, accurate satellite data, streamlining monitoring and management, and enhancing energy policy and development decisions.

Frequently Asked Questions (FAQ)

What is SolarMap and who benefits from it?

SolarMap is an innovative platform developed by AWS and the Ministry of Electricity and Water Affairs in Bahrain. It integrates a map and solar panel calculator to support efficient solar energy management and policy development. Beneficiaries include MEWA, energy providers, policy-makers, and citizens by promoting efficient renewable energy management and supporting sustainable development goals.

How does SolarMap address existing challenges?

SolarMap solves the time-consuming and costly process of manually mapping and managing solar installations by offering real-time, accurate satellite data. This streamlines monitoring and management, enhancing decision-making for energy policies and development. SolarMap integrates with Bahrain's MEWA rates, providing accurate cost estimations and ROI calculations, making it the only Bahraini renewable energy calculator of its kind.

How can stakeholders access and utilize SolarMap?

Stakeholders can access SolarMap through a dedicated web platform. The platform features interactive maps and analytics tools for comprehensive solar installation data. Homeowners and businesses can use the portal to estimate solar panel requirements, costs, and connect with recommended contractors, while administrators can visualize and manage installations across Bahrain.